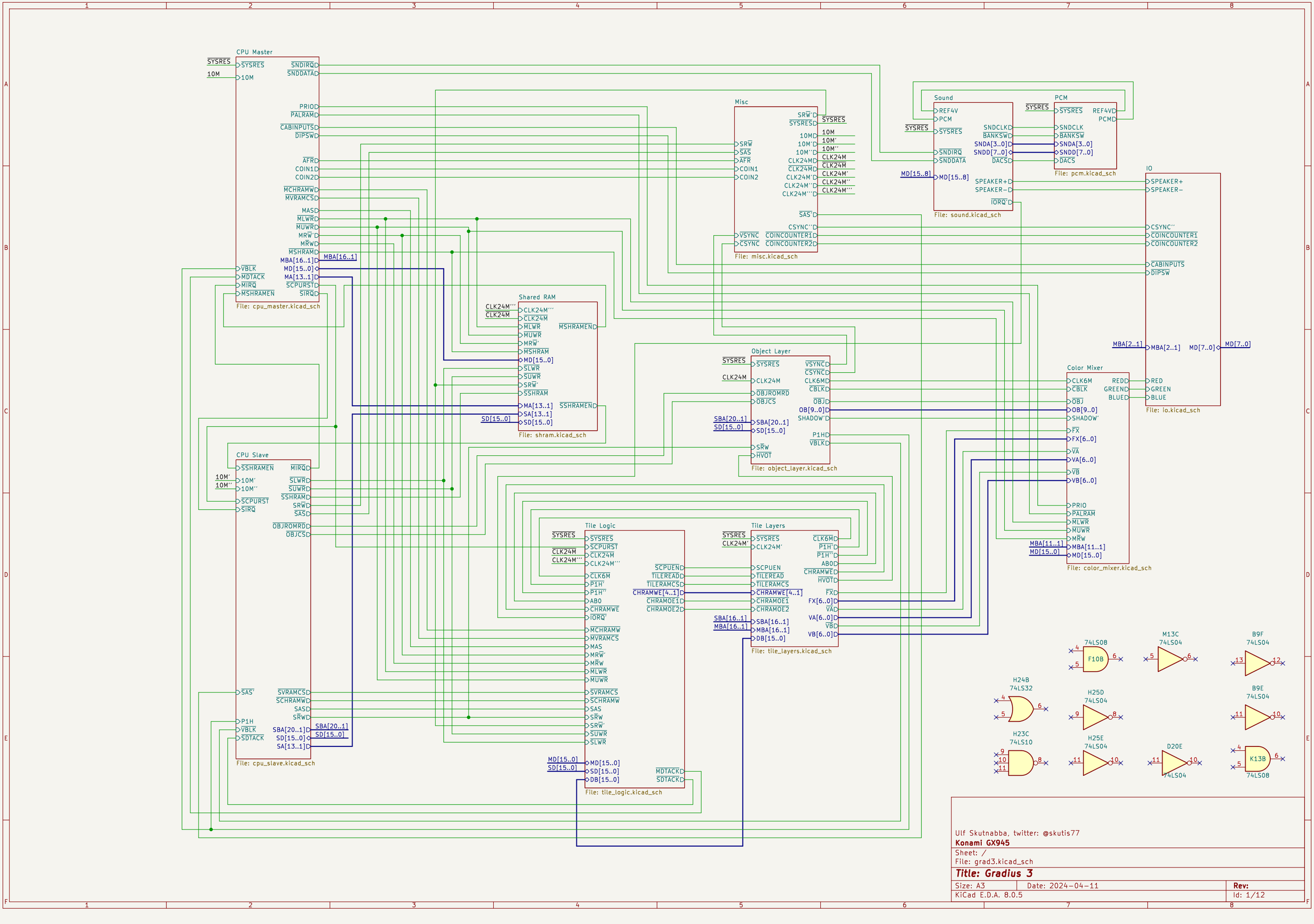
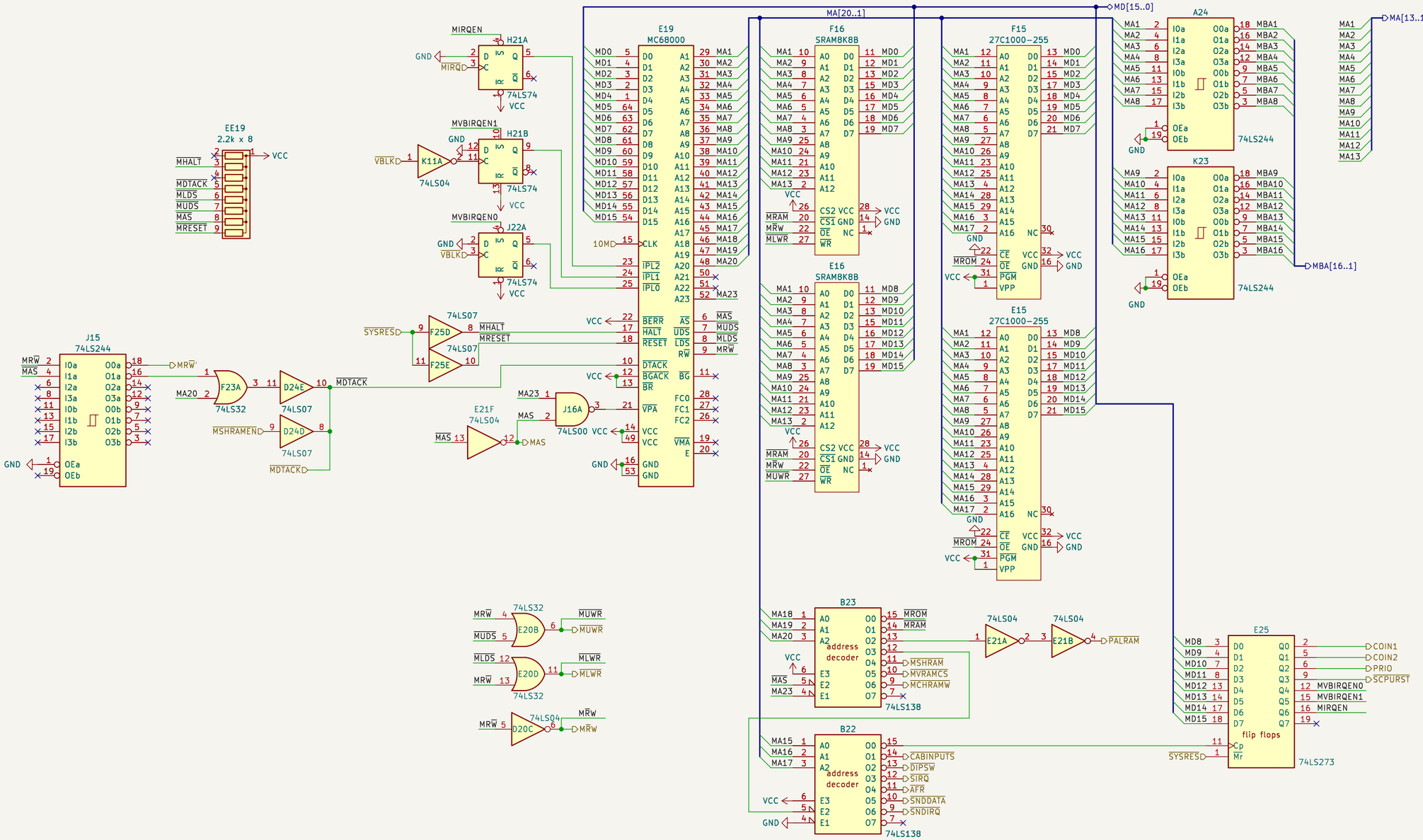

CPU Master
SYSRES
10M
SNDIRQ
SNDDATA
PRIO
PALRAM
CABINPUTS
DIPSW
AFR
COIN1
COIN2
MCHRAMW
MVRAMCS
MAS
MLWR
MUWR
MRW'
MRW
MSHRAM
MBA[16..1]
MD[15..0]
MA[13..1]
SCPURST
SIRQ
VBLK
MDTACK
MIRQ
MSHRAMEN
File: cpu_master.kicad_sch
Misc
SRW'
SYSRES'
10M
10M'
10M''
CLK24M
CLK24M'
CLK24M''
CLK24M'''
SRW
SAS
AFR
COIN1
COIN2
SAS'
CSYNC''
VSYNC
CSYNC
COINCOUNTER1
COINCOUNTER2
File: misc.kicad_sch
Sound
REF4V
PCM
SYSRES
SNDIRQ
SNDDATA
SNDCLK
BANKSW
SNDA[3..0]
SNDD[7..0]
DACS
MD[15..8]
SPEAKER+
SPEAKER-
IORQ'
File: sound.kicad_sch
PCM
SYSRES
SNDCLK
BANKSW
SNDA[3..0]
SNDD[7..0]
DACS
REF4V
PCM
File: pcm.kicad_sch
IO
SPEAKER+
SPEAKER-
CSYNC''
COINCOUNTER1
COINCOUNTER2
CABINPUTS
DIPSW
MBA[2..1]
MBA[2..1]
MD[7..0]
MD[7..0]
RED
GREEN
BLUE
File: io.kicad_sch
Shared RAM
CLK24M'''
CLK24M
MLWR
MUWR
MRW'
MSHRAM
MD[15..0]
SLWR
SUWR
SRW'
SSHRAM
MSHRAMEN
SSHRAMEN
MA[13..1]
SA[13..1]
SD[15..0]
File: shram.kicad_sch
Object Layer
SYSRES
CLK24M
OBJROMRD
OBJCS
SBA[20..1]
SD[15..0]
SRW
HVOT
VSYNCD
CSYNCD
CLK6MD
CBLKD
OBJD
OB[9..0]D
SHADOW'D
P1HD
VBLKD
File: object_layer.kicad_sch
Color Mixer
CLK6M
CBLK
OBJ
OB[9..0]
SHADOW'
FX
FX[6..0]
VA
VA[6..0]
VB
VB[6..0]
PRIO
PALRAM
MLWR
MUWR
MRW
MBA[11..1]
MD[15..0]
REDD
GREEND
BLUED
MBA[11..1]
MD[15..0]
File: color_mixer.kicad_sch
CPU Slave
SSHRAMEN
10M'
10M''
SCPURST
SIRQ
MIRQ
SLWRD
SUWRD
SSHRAM
SRW
SAS
OBJROMRD
OBJCS
SAS'
P1H
VBLK
SDTACK
SVRAMCS
SCHRAMW
SAS
SRW
SBA[20..1]
SD[15..0]
SA[13..1]
SBA[20..1]
SD[15..0]
File: cpu_slave.kicad_sch
Tile Logic
SYSRES
CLK24M
CLK24M'''
SCPURST
CLK24M
CLK24M'''
CLK6M
P1H'
P1H''
ABO
CHRAMWE
IORQ'
MCHRAMW
MVRAMCS
MAS
MRW'
MRW
MLWR
MUWR
SVRAMCS
SCHRAMW
SAS
SRW
SRW'
SUWR
SLWR
MD[15..0]
SD[15..0]
DB[15..0]
SCPUEN
TILEREAD
TILERAMCS
CHRAMWE[4..1]
CHRAMOE1
CHRAMOE2
MDTACKD
SDTACKD
MD[15..0]
SD[15..0]
DB[15..0]
File: tile_logic.kicad_sch
Tile Layers
SYSRES
CLK24M'
SCPUEN
TILEREAD
TILERAMCS
CHRAMWE[4..1]
CHRAMOE1
CHRAMOE2
SBA[16..1]
MBA[16..1]
DB[15..0]
CLK6MD
P1H'D
P1H''D
ABOD
CHRAMWED
HVOTD
FXD
FX[6..0]
VAD
VA[6..0]
VBD
VB[6..0]
SBA[16..1]
MBA[16..1]
File: tile_layers.kicad_sch
74LS08
F10B
M13C
74LS04
B9F
74LS04
H24B
74LS32
H25D
74LS04
B9E
74LS04
H23C
74LS10
H25E
74LS04
D20E
74LS04
K13B
74LS08
Ulf Skutnabba, twitter: @skutis77
Konami GX945
Sheet: /
File: grad3.kicad_sch
Title: Gradius 3
Size: A3
Date: 2024-04-11
Rev:
KiCad E.D.A. 8.0.5
Id: 1/12



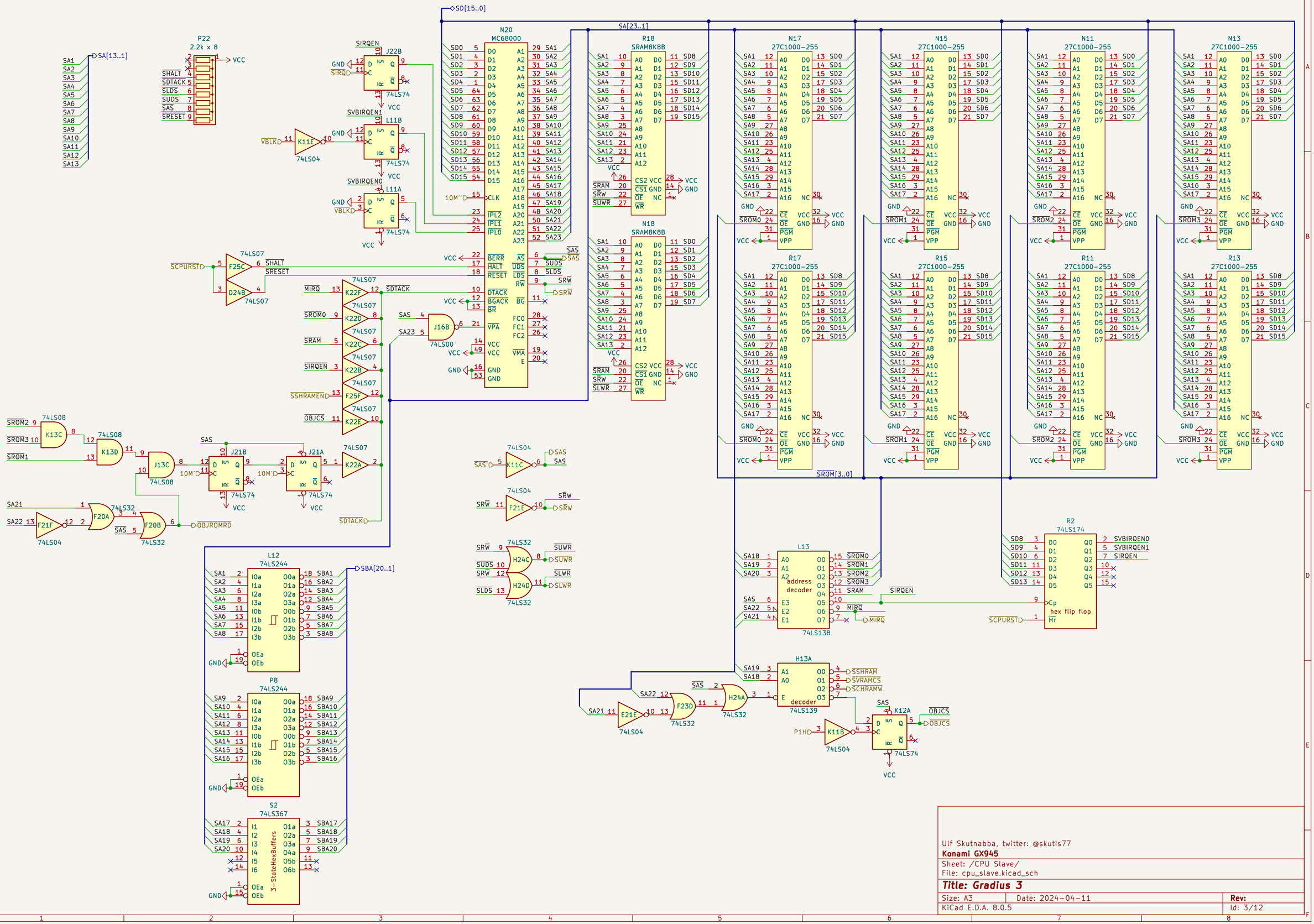
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Sheet: /CPU Master/
File: cpu_master.kicad_sch

Title: Gradius 3

Size: A3	Date: 2024-04-11	**Rev:**
KiCad E.D.A. 8.0.5		Id: 2/12



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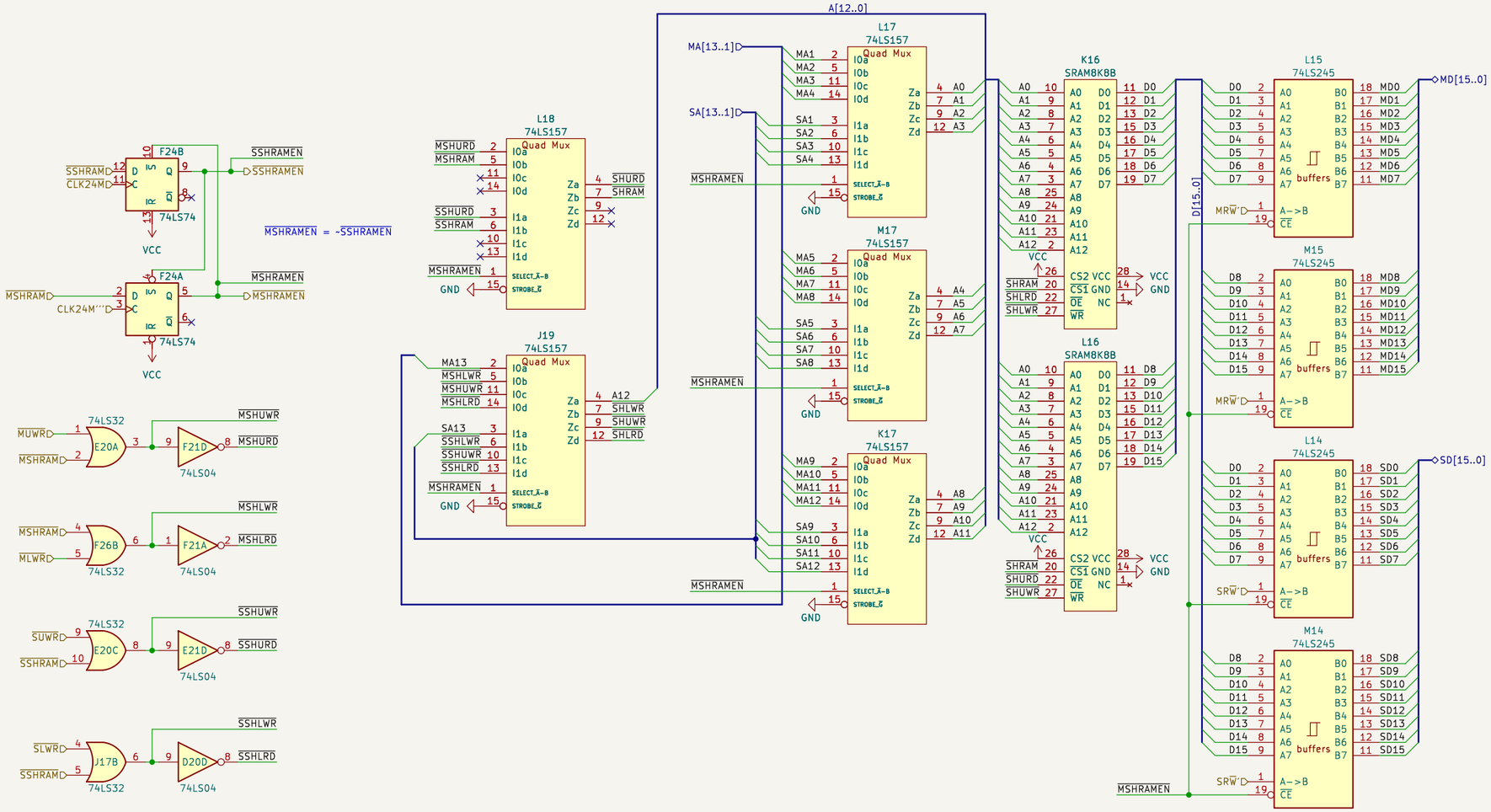
Konami GX945

Sheet: /CPU Slave/

File: cpu_slave.kicad_sch

Title: Gradius 3

Size: A3	Date: 2024-04-11	**Rev:**
KiCad E.D.A. 8.0.5		Id: 3/12



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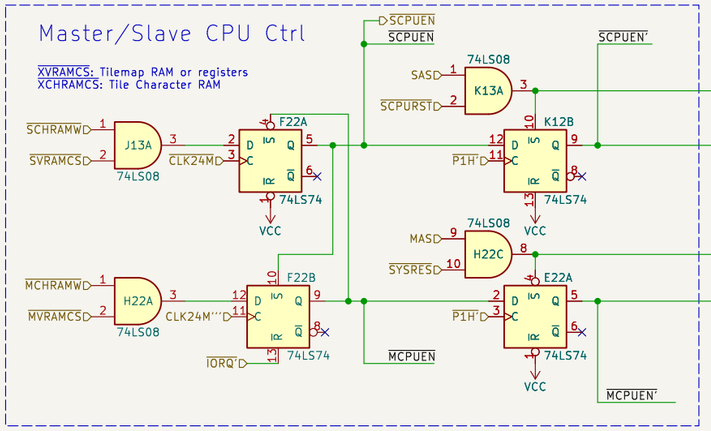
Sheet: /Shared RAM/
File: shram.kicad_sch

Title: Gradius 3

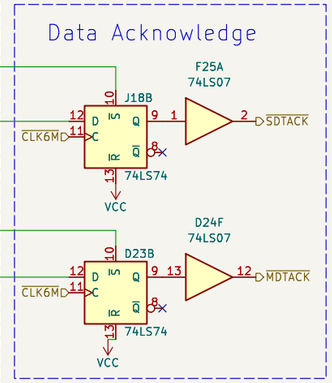
Size: A3	Date: 2024-04-11	**Rev:**
KiCad E.D.A. 8.0.5		Id: 4/12

Master/Slave CPU Ctrl

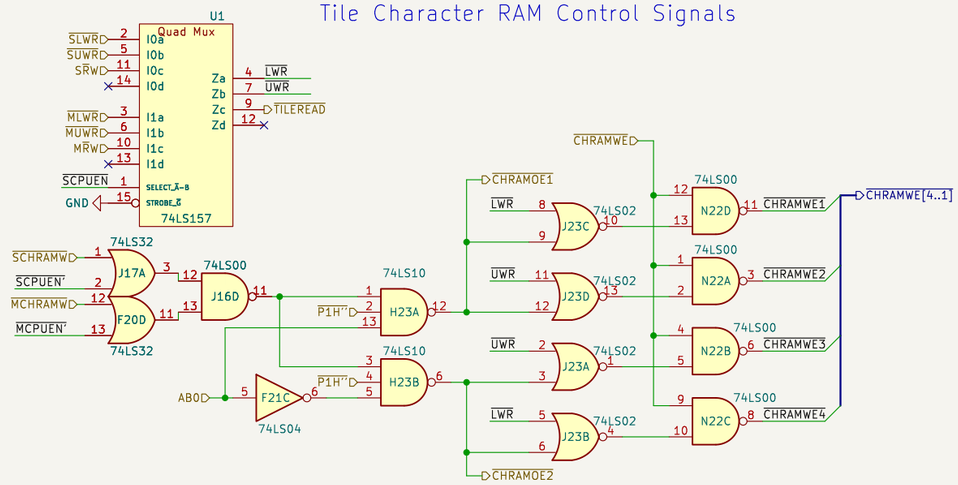
XVRAMCS: Tilemap RAM or registers
XCHRAMCS: Tile Character RAM



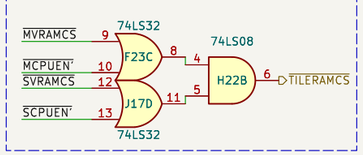
Data Acknowledge



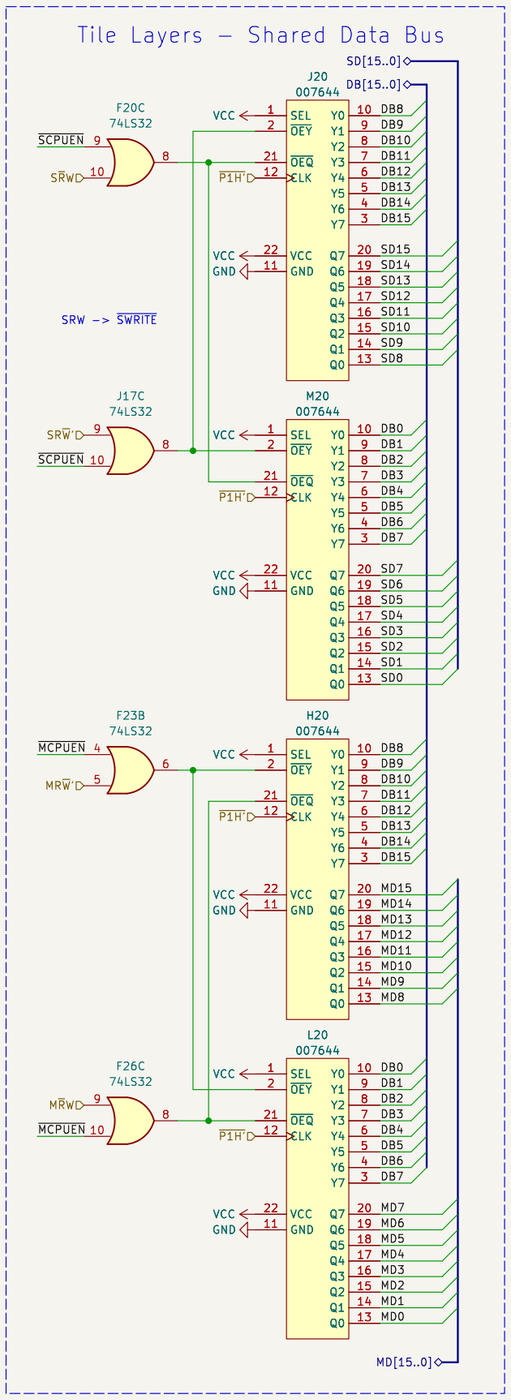
Tile Character RAM Control Signals



Tile RAM and Registers



Tile Layers — Shared Data Bus



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Sheet: /Tile Logic/
File: tile_logic.kicad_sch

Title: Gradius 3

Size: A3	Date: 2024-04-11	**Rev:**
KiCad E.D.A. 8.0.5		Id: 5/12

Character code RAM

TILERAMCS can write and read both to tilemap RAM and character RAM

Tilemaps RAM: Fix, Layer 1 and 2

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Konami GX945

Sheet: /Tile Layers/

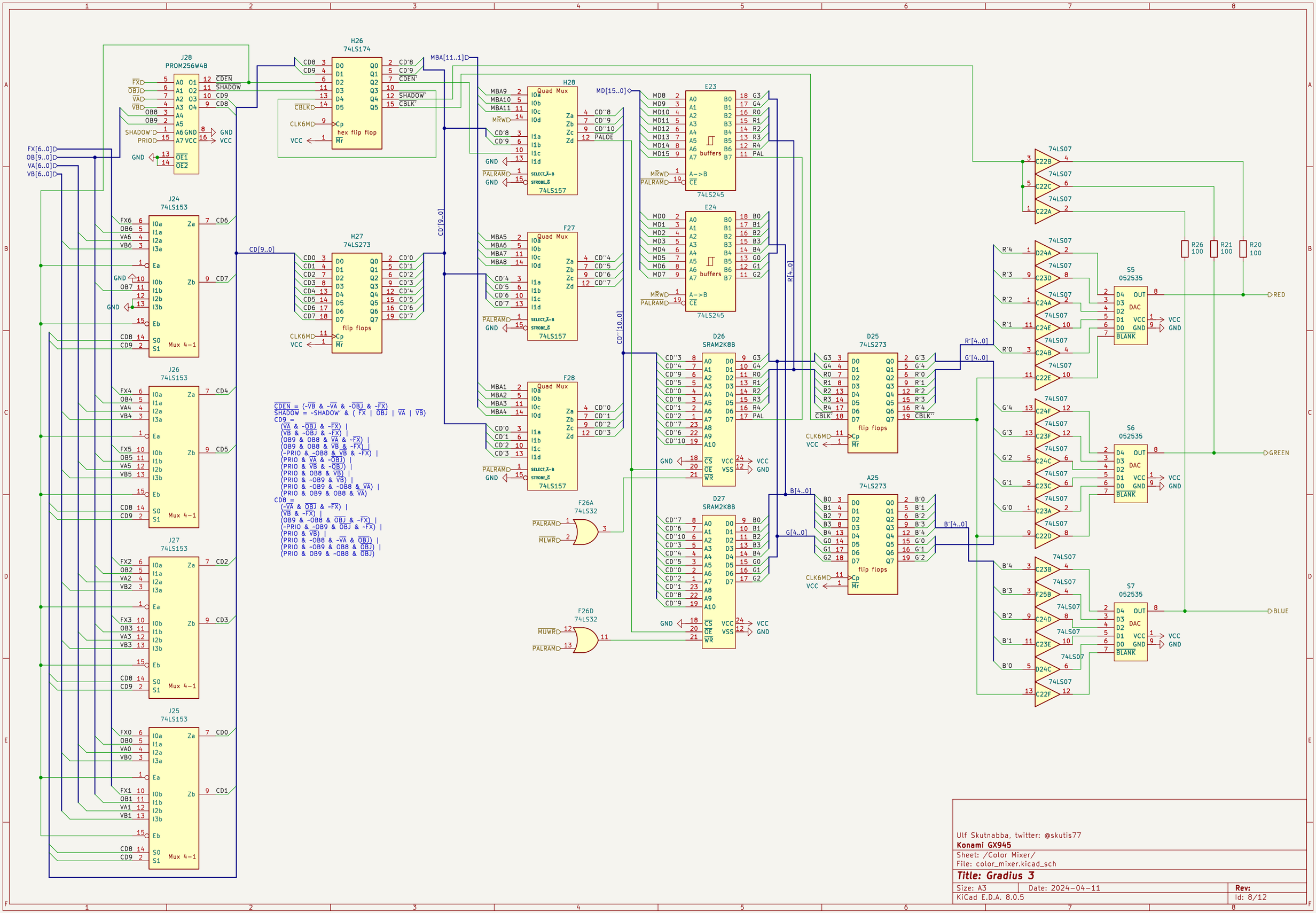
File: tile_layers.kicad_sch

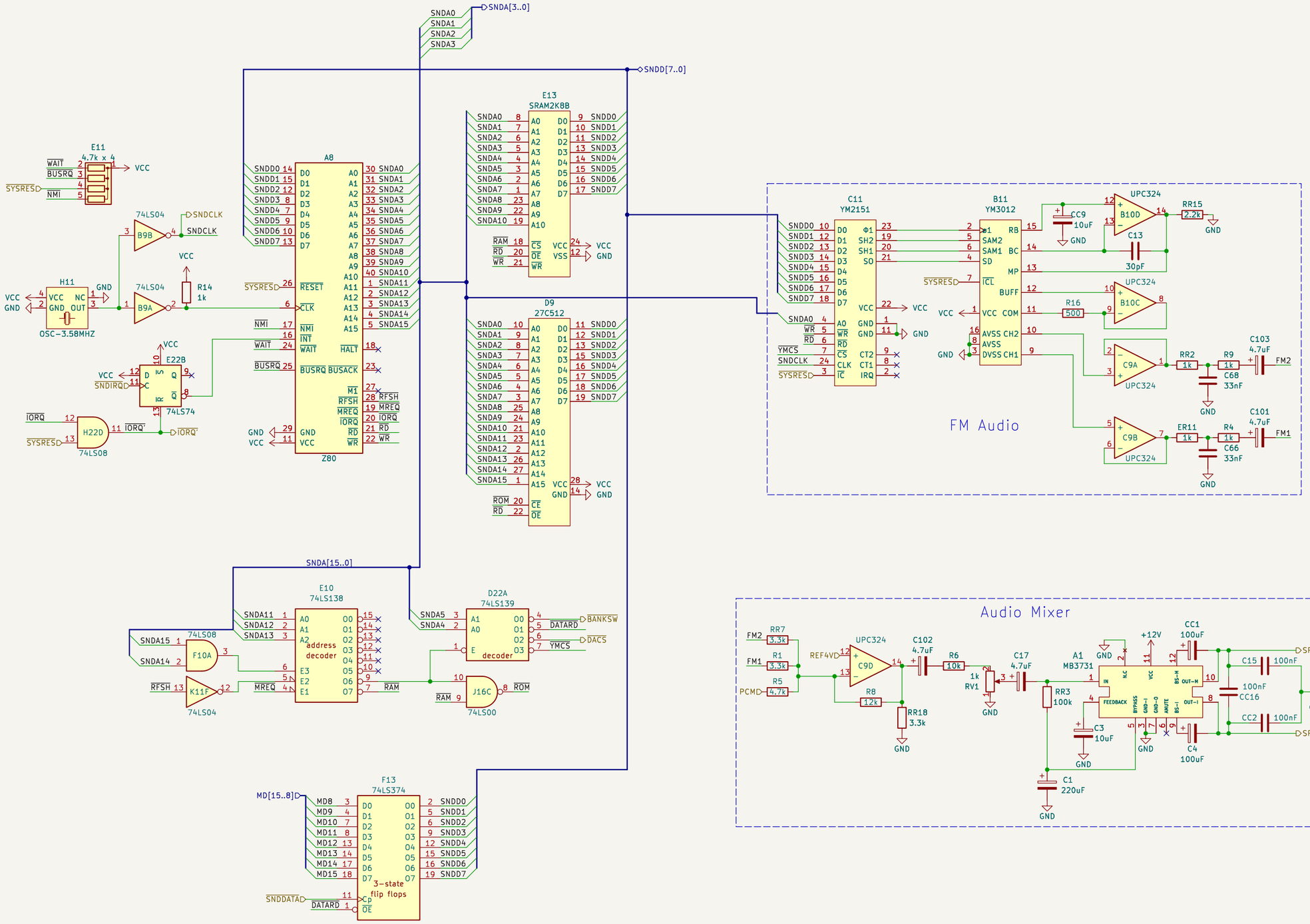
Title: Gradius 3

Size: A3 | Date: 2024-04-11 | Rev:

KiCad E.D.A. 8.0.5 | Id: 6/12

R7
051960
SBA[20..1]
CA[17..0]
OBJROMRD
OC[7..1]
HP[8..0]
OA[9..0]
OD[7..0]
{OBJCTRL}
P7
74LS157
Quad Mux
P5
74LS157
Quad Mux
P4
74LS157
Quad Mux
P3
74LS157
Quad Mux
P2
74LS157
Quad Mux
R3
SRAM2K8B
A[18..0]
H13B
74LS139
decoder
H6
K6
M6
E6
H8
K8
M8
E8
27C1000
CD[37..0]
SD[15..0]
H10
74LS244
J10
74LS244
L10
74LS244
M10
74LS244
P6C
74LS32
P6D
74LS32
P6A
74LS32
P6B
74LS32
K11D
74LS04
J13D
74LS08
J13B
74LS08
OROM[1..0]
H3
23C4100
L3
23C4100
D4
051937
OB[9..0]
Goes to optional
ROM Modules
F10C
74LS08
F10D
74LS08
Not used on gradius 3
Ulf Skutnabba, twitter: @skutis77
Konami GX945
Sheet: /Object Layer/
File: object_layer.kicad_sch
Title: Gradius 3
Size: A3
Date: 2024-04-11
KiCad E.D.A. 8.0.5
Rev:
Id: 7/12


J28
PROM256W4B
H26
74LS174
MBA[11..1]
MD[15..0]
H28
Quad Mux
E23
74LS245
E24
74LS245
H27
74LS273
CD[9..0]
CD''[9..0]
F27
Quad Mux
F28
Quad Mux
D26
SRAM2K8B
D25
74LS273
A25
74LS273
D27
SRAM2K8B
F26A
74LS32
F26D
74LS32
J24
74LS153
J26
74LS153
J27
74LS153
J25
74LS153
R[4..0]
B[4..0]
G[4..0]
R'[4..0]
G'[4..0]
B'[4..0]
S5
052535
S6
052535
S7
052535
RED
GREEN
BLUE
R26 100
R21 100
R20 100
CDEN = (~VB & ~VA & ~OBJ & ~FX)
SHADOW = ~SHADOW' & (FX | OBJ | VA | VB)
CD9 =
(VA & ~OBJ & ~FX) |
(VB & ~OBJ & ~FX) |
(OB9 & OB8 & VA & ~FX) |
(OB9 & OB8 & VB & ~FX) |
(~PRIO & ~OB8 & VB & ~FX) |
(PRIO & VA & ~OBJ) |
(PRIO & VB & ~OBJ) |
(PRIO & OB8 & VB) |
(PRIO & ~OB9 & VB) |
(PRIO & ~OB9 & ~OB8 & VA) |
(PRIO & OB9 & OB8 & VA)
CD8 =
(~VA & OBJ & ~FX) |
(VB & ~FX) |
(OB9 & ~OB8 & OBJ & ~FX) |
(~PRIO & ~OB9 & OBJ & ~FX) |
(PRIO & VB) |
(PRIO & ~OB8 & ~VA & OBJ) |
(PRIO & ~OB9 & OB8 & OBJ) |
(PRIO & OB9 & ~OB8 & OBJ)
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Konami GX945
Sheet: /Color Mixer/
File: color_mixer.kicad_sch
Title: Gradius 3
Size: A3
Date: 2024-04-11
Rev:
KiCad E.D.A. 8.0.5
Id: 8/12


SNDA[3..0]
SNDD[7..0]
SNDA[15..0]
MD[15..8]
E11
4.7k x 4
VCC
SYSRES
WAIT
BUSRQ
NMI
74LS04
B9B
B9A
SNDCLK
H11
OSC-3.58MHZ
R14
1k
E22B
74LS74
SNDIRQ
H22D
74LS08
IORQ
A8
Z80
RESET
CLK
NMI
INT
WAIT
HALT
BUSRQ
BUSACK
M1
RFSH
MREQ
IORQ
RD
WR
E13
SRAM2K8B
RAM
D9
27C512
ROM
E10
74LS138
address decoder
F10A
K11F
D22A
74LS139
decoder
BANKSW
DATARD
DACS
YMCS
J16C
74LS00
F13
74LS374
3-state flip flops
SNDDATA
C11
YM2151
B11
YM3012
UPC324
B10D
B10C
C9A
C9B
CC9
10uF
C13
30pF
R16
500
RR15
2.2k
RR2
RR9
C68
33nF
C103
4.7uF
FM2
ER11
R4
C66
C101
FM1
FM Audio
Audio Mixer
RR7
3.3k
R1
R5
4.7k
PCM
FM2
FM1
C9D
REF4V
R8
12k
RR18
C102
R6
10k
RV1
C17
RR3
100k
A1
MB3731
CC1
C15
CC16
CC2
C3
C4
C1
220uF
SPEAKER+
SPEAKER-
+12V
Ulf Skutnabba, twitter: @skutis77
Konami GX945
Sheet: /Sound/
File: sound.kicad_sch
Title: Gradius 3
Size: A3
Date: 2024-04-11
Rev:
KiCad E.D.A. 8.0.5
Id: 9/12

B15 and B18 do not follow the standard Mask ROM behaviour. The BYTE mode select must be selected from factory and pin 31 is probably not connected.

BANK A

Mounted on all releases except (Japan, program code S, split)

B15
23C4100

BANK B

Not mounted on any releases

B18
23C4100

Mounted on Gradius III (Japan, program code S, split)

C14
27C1000

C16
27C1000

C18
27C1000

C20
27C1000

PCM ROM Data Bank Select

SNDD[7..0]

D21
74LS174

B21
74LS157
Quad Mux

C21
74LS174

74LS04

D22B
74LS139
decoder

BANKSW
SYSRES
SNDCLK4
SNDCLK2
DABNK0
DABNK1
DABNKA0
DABNKA1
DABNKB0
DABNKB1
GND

SNDCLK 3.579545MHz
SNDCLK2 1.789773MHz
SNDCLK4 0.894886MHz

SNDA[3..0]
SNDA0
B9D
74LS04
B9C
74LS04
D23A
74LS74
SYSRES
SNDCLK2
SNDCLK4

AS[16..0]
DS[7..0]

C2
007232

D7
74LS273
flip flops

D10
4.7k x 8
VCC

Q[0..7]

EE6
4066

D8
KONAMI_007341

C9C
UPC324
REF4V
R10
10k
C63
4.7uF
PCM

D6
4066

DD7
KONAMI_007341

UPC324
B10B
B10A
B10E
C9E

R19
3.3k
RR17
3.3k
+12V
CC21
4.7uF
CC14
4.7uF

B7
4.7k x 4
B8
4.7k x 4
C64
4.7uF
REF4V

RR13
20k
RR12
20k
C10
470pF
C105
470pF
C104
4.7uF
C100
4.7uF

007340
A3
GND B3

The values to the right are the capacitors on the PCB. C10X are not visible on the PCB. They are values set to ease debugging.

C104
C103
C102
C10
C101
C100
C65
C67
C105

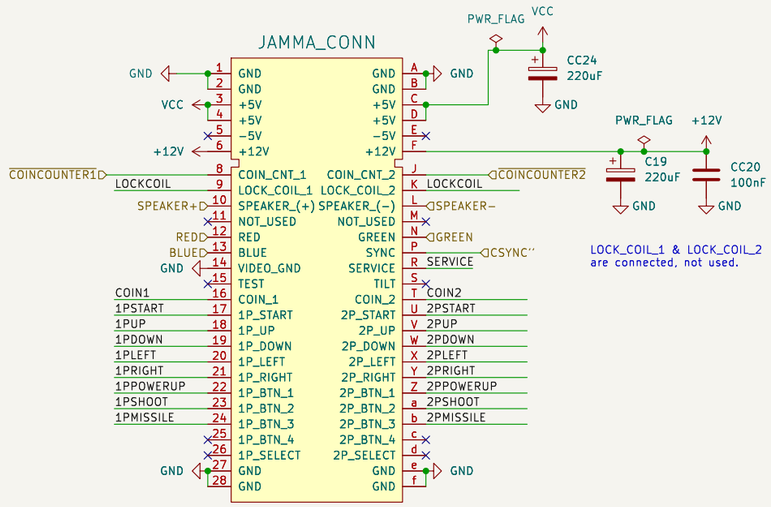
Ulf Skutnabba, twitter: @skutis77

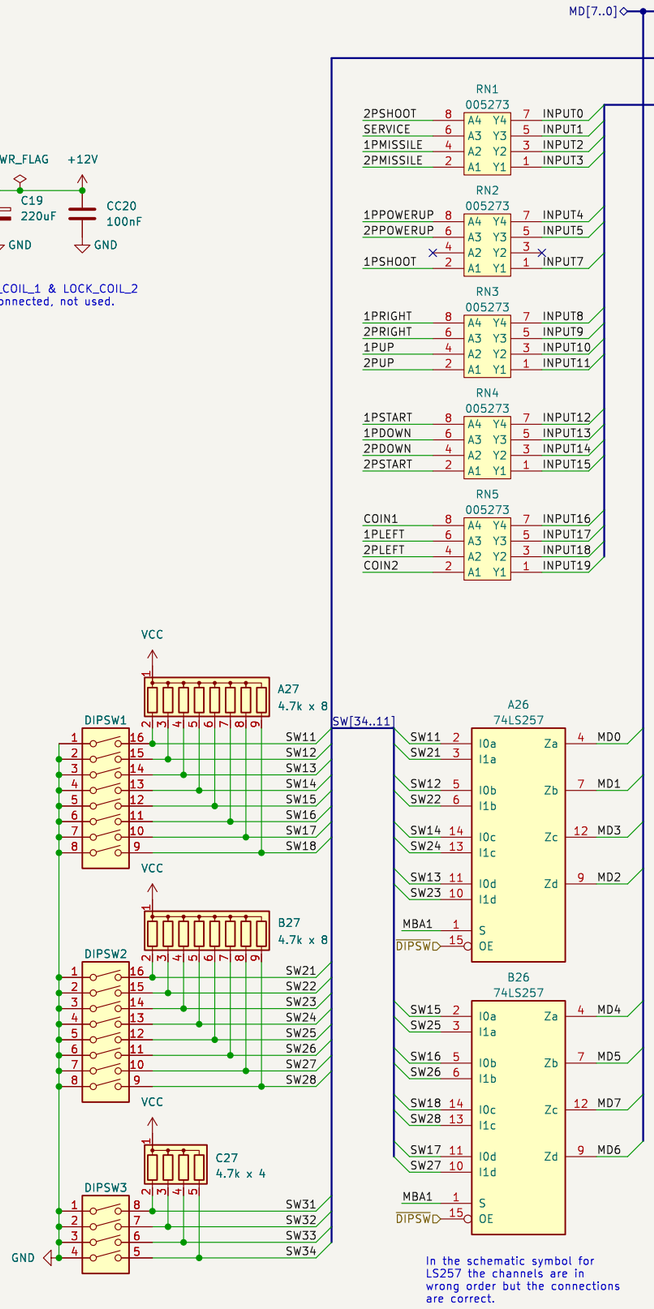
Konami GX945

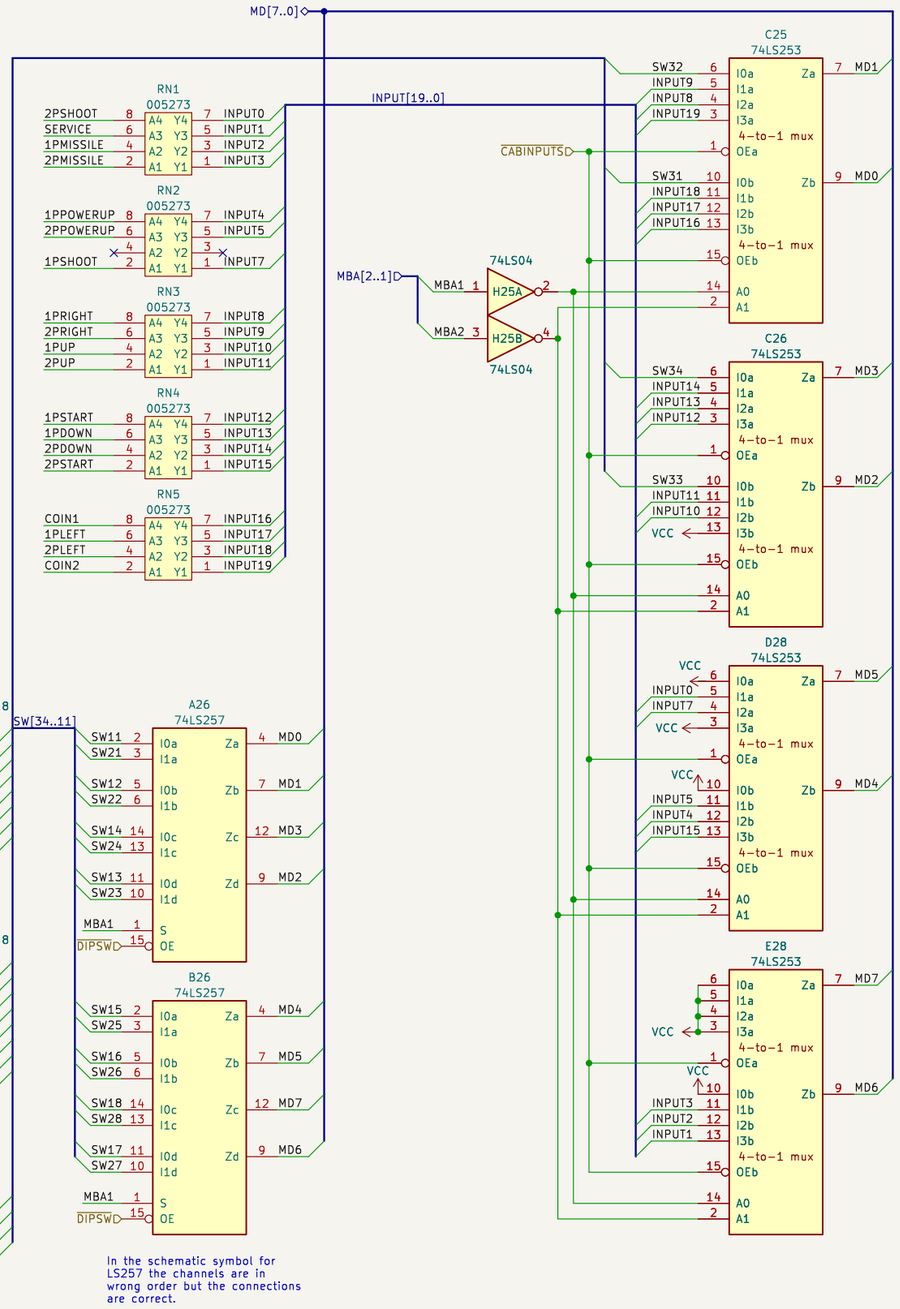
Sheet: /PCM/
File: pcm.kicad_sch

Title: Gradius 3

Size: A3	Date: 2024-04-11	**Rev:**
KiCad E.D.A. 8.0.5		Id: 10/12







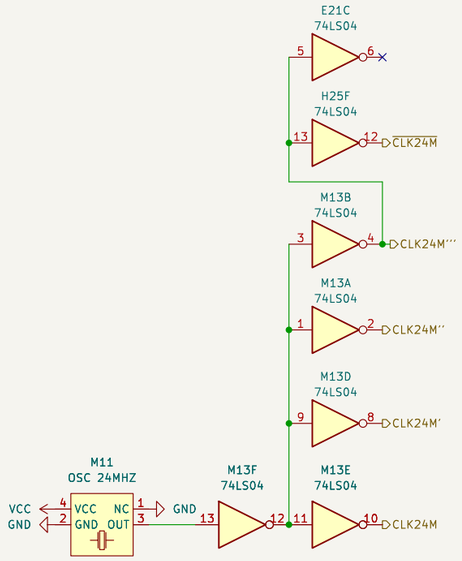
Ulf Skutnabba, twitter: @skutis77

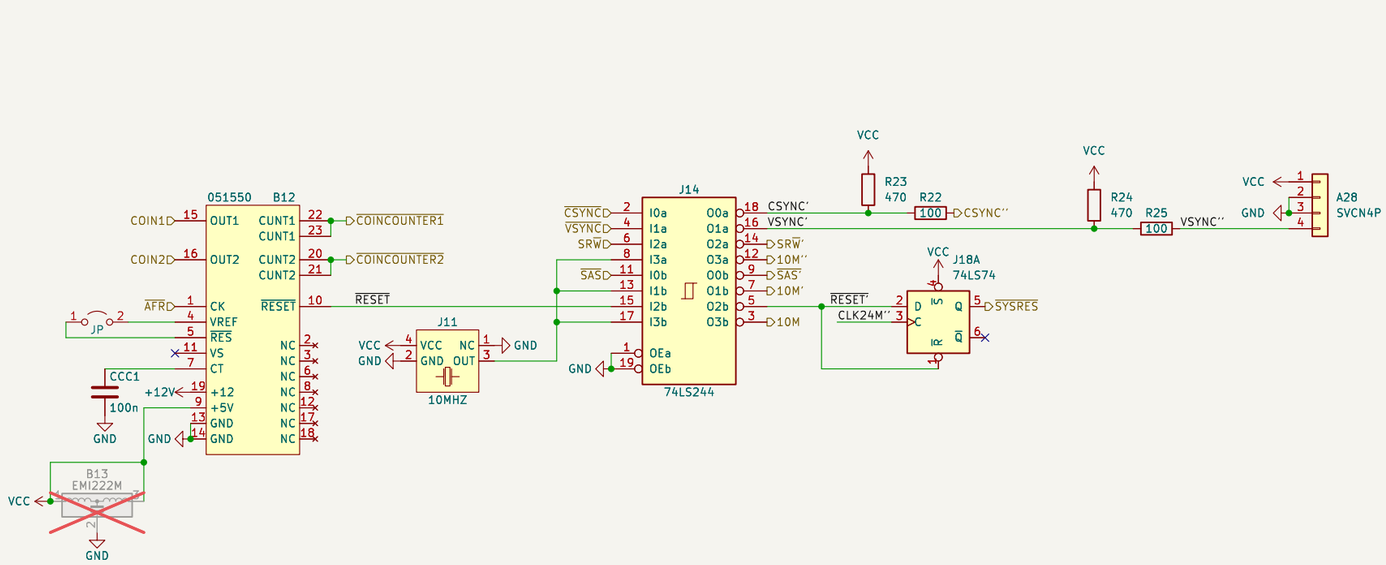
Konami GX945

Sheet: /IO/
File: io.kicad_sch

Title: Gradius 3

Size: A3	Date: 2024-04-11	**Rev:**
KiCad E.D.A. 8.0.5		Id: 11/12





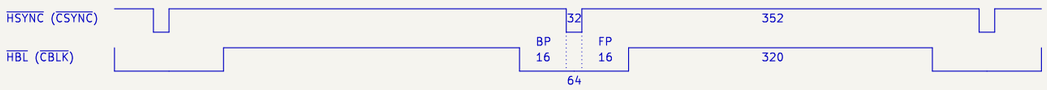
Horizontal and vertical synch timing diagrams

The pixel clock is derived from the 24MHz oscillator.
Pixel clock OVCK: f = 24MHz / 4 = 6MHz

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of the pixel clock.

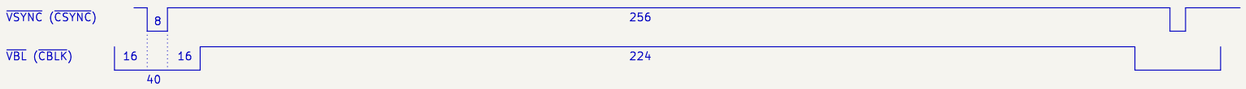
The signals have been measured at the output of the
graphic chips.

If horizontal blanking is measured at the RGB DACs, the blanking
is delayed 2 pixel clocks relative to composite sync. This
gives BP = 14 and FP = 18.



HSYNC and HBL
Frequency f = 6MHz / 384 = 15.625kHz.
Period T = 1/f = 64us.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency f = 15.625kHz / 264 = 59.1856Hz
Period T = 1/f = 1 / 59.1856Hz = 16.896ms

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Konami GX945

Sheet: /Misc/
File: misc.kicad_sch

Title: Gradius 3

Size: A3	Date: 2024-04-11	**Rev:**
KiCad E.D.A. 8.0.5		Id: 12/12